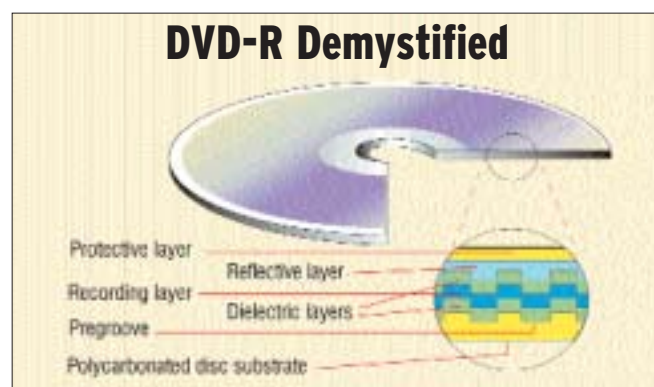
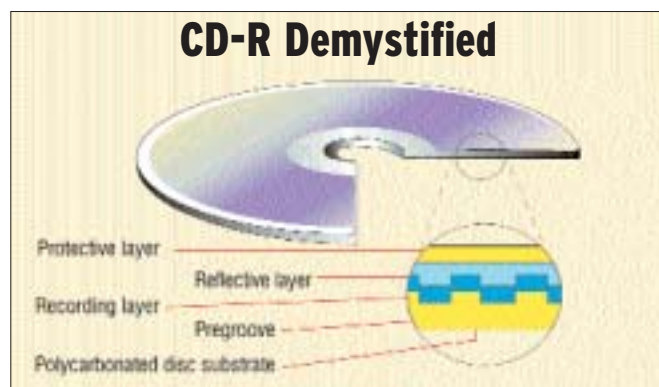




The Future of Storage Devices

Secondary storage devices have evolved a great deal from the slow, cumbersome and low-capacity floppy drives to the latest DVD-Writers capable of storing gigabytes of data. The future seems very bright, with new inventions and newer standards being formed to cope with the growing storage needs on multiple platforms.

CD-Writers have theoretically reached a saturation point from where further upgrading seems impossible. Many vendors like Plextor have invented new ways of crunching more data into a normal 700 MB disk but they are plagued with incompatibility issues with existing CD-ROM drives, which are incapable of reading such 'oversized' disks.

On the other hand, DVD-Writers are still in a nascent stage and hence more promising than other devices. Currently available DVD-R can store about 4 GB of data, but the newer Blue Ray disks—which have just appeared on store shelves—can store a mammoth 23 GB data on a single disk.

Burn-Proof Technology

Buffer underruns occur when the computer fails to provide data to the CD-Writer in a continuous stream. As a result, the record-

ing fails and the media becomes unusable. This can happen when attempting to burn a CD-R while performing other tasks, or when recording from a relatively 'slow' source to a 'fast' target.

In the past, you needed to shut down all the other activities running on your PC and wait till the CD-Writer finished writing the CD. With the advent of Burn-Proof technology, you no longer need to continuously monitor the buffer bar. Introduced by Sanyo, CD-Writers featuring this technology have become intelligent enough to synchronise, wait and then burn the data.

Using this technology, recording is paused at a specific location when an impending buffer-underrun condition is detected. The writing process resumes only when the buffer has been sufficiently replenished and after repositioning the CD-Writer's optical pickup mechanism to the appropriate sector.

Most major drive manufactures either license similar technology or have developed their own variants. Mitsumi and LG Electronics use OAK Technology's ExaLink system, Yamaha does the same but under the brand name SafeBurn. Acer Communications refers to their technology as Seamless Link and Ricoh calls theirs JustLink.



External v/s Internal Writers

Choosing between an external and an internal drive depends primarily on your requirements.

- Internal drives are usually cheaper. You can choose these if you are a home user and not always on the move.
- Till now external writers were slower compared to internal writers due to their interface but with the arrival of FireWire and USB 2.0, this is no longer a problem. Drives based on the SCSI interface were traditionally the fastest.
- The greatest advantage of an external USB CD-Writer is that it can be hot-plugged to the PC (that is, there's no need to shut down and restart).
- External writers come with the added baggage of SCSI/FireWire/USB 2.0 cards, which adds to the budget.
- Internal CD-Writers tend to hog a great deal more of the system's resources.
- Buffer underruns used to be a big problem with internal drives, but that's been solved with the advent of buffer underrun protection technology.

Hybrid DVDs

Nine leading consumer electronics manufacturers have jointly established the basic specifications for a next-generation large-capacity optical disk video recording format called Blue Ray. These DVDs can hold six times as much data as current standard disks. The new format can store more than 13 hours of film, compared with the current limit of 133 minutes.

High Density-DVD is an enhanced form of current DVD format. Blue lasers have a much shorter wavelength than the red lasers used in current DVD players, allowing for much greater disk capacity. The new high definition format allows a capacity of 4.7-15 GB on a single-sided, single-layered disk, and a capacity of 8.5-30 GB on a single-sided, double-layered disk. Some feel this is a cost-effective upgrade path for media vendors—and a backward-compatible solution for those who've already built DVD libraries.

Current DVD players use 650 nanometre (nm) red lasers, while CD players use 780 nm diodes. It won't be possible to play Blue Ray disks with existing DVD and CD players, since these don't contain 405 nm lasers. Blue Ray drives will need to contain 780, 650 and 405 nm lasers in order to be compatible with CD and DVD disks.